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A new controller design method for an electric power steering system based on a target steering torque feedback controller

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ABSTRACT

This paper describes a novel electric power steering (EPS) system controller that distinguishes the steering feel design problem from the system stability and control performance problem and provides a design methodology for each problem. The suggested controller, called the *target steering torque feedback controller*, consists of two modules: the *steering feel generation logic* module defines the target steering torque that the driver should feel under a given driving condition, and the *steering torque feedback controller* module controls the assist motor so that the driver actually feels the defined target torque. By disassociating all nonlinear elements related to steering feel tuning from the feedback controller, the inherent difficulties of the conventional EPS controller in terms of the coupling of system stability analysis and steering feel design are overcome. Rack-force-based steering feel generation logic that creates a familiar steering feel for drivers while transmitting the road condition information is suggested and design criteria of the steering torque feedback controller in the frequency domain are proposed to ensure robust stability, tracking performance and noise attenuation. In addition, a disturbance observer is applied to the feedback controller to compensate for the effect of disturbances and to improve tracking performance. The proposed controller design method is verified by conducting computer simulations and a vehicle test.

1. Introduction

The electric power steering (EPS) system is an essential system that provides a light and comfortable steering feel to the driver by using an electric motor to assist the torque required for steering, as shown in Fig. 1

An important component of the overall system performance of an EPS system is the design of the controller, which determines the amount of assist motor torque. If the controller is not designed correctly, the steering system may become unstable, causing vibration or discomfort to the driver.

A conventional EPS controller generally consists of three components: (1) the torque map, (2) the stabilization logic and (3) the additional logic for detailed tuning of the steering characteristics, as shown in Fig. 2.

The torque map is a nonlinear map that determines the magnitude of the assist motor torque based on the driver's steering torque measured by the torque sensor and vehicle speed information. Because the assist torque of motor is mostly determined by the torque map, the torque map is the factor that most affects the driver's steering feel.

A steep torque map slope at low-speed or parking conditions which highly amplifies the driver's steering torque and the nonlinearity of the torque map makes the system unstable (Lee, Kim and Kim, 2018; Zaremba, Liubakka, & Stuntz, 1998). Therefore, it is necessary to design stabilization logic to ensure that the steering feel defined by the designed torque map can be transmitted to the driver without system instability or unnecessary vibration.

Since the torque map represents only static characteristics of the steering system, it does not reflect all the steering characteristics of driving. For a real steering system, additional logics are required to tune dynamic steering characteristics such as damping, inertia and friction. Generally, torque calculated from these additional logics is added to the main assist torque determined by the torque map and stabilization logic, as shown in Fig. 2.

Although studies on various issues have been conducted using the conventional EPS controller structure shown in Fig. 2, these previous studies have inherent limitations because a nonlinear torque map and additional logics are included inside the controller.

First, the nonlinear torque map of the controller makes the entire controller and system nonlinear, which makes it difficult to analyze the stability and performance of the system. The characteristics of the system in the frequency domain are important for delivering the

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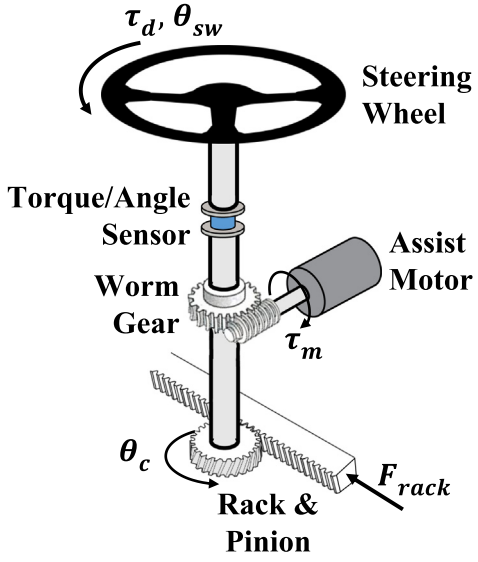


Fig. 1. Schematic structure of the electric power steering system.

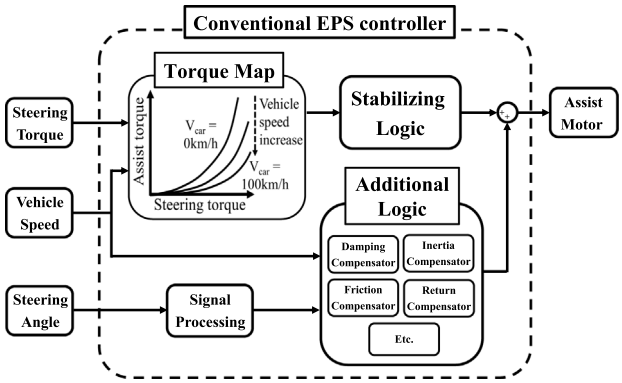


Fig. 2. Typical structure of a conventional EPS controller.

proper steering torque characteristics to the driver. Therefore, many previous studies have applied linear control theory based on an approximately linear system that approximates nonlinear torque map as simple constant amplifying gains (Chabaan, 2001; Chen, Yang, Chen, & Zhou, 2008; Chitu et al., 2013; Kurishige, Nishihara, & Kumamoto, 2010; Lee, Kim et al., 2018; Mehrabi, Azad, & McPhee, 2011; Zaremba et al., 1998). However, since the linearly approximated system model has different characteristics from original systems, the performance of the proposed controller is not satisfactory for real applications. There have been other attempts to apply nonlinear controllers such as the sliding mode controller (Marouf, Djemai, Sentouh, & Pudlo, 2012). However, the suggested controller has difficulty in suppressing undesired vibration and ensuring robustness to external disturbances, which are the most important requirements of EPS controllers.

Furthermore, additional logics for tuning the driver's steering characteristics makes it even more difficult to analyze the stability and to design the controller. Since the additional logics utilize various state variables of the vehicle, the complexity of the controller structure is significantly increased. Therefore, it is almost impossible to analyze the stability of the system considering all such logics. Consequently, most studies on EPS controllers ignore the additional logic when they design the controller and expect that the robustness of the controller can handle the effects of the additional logic (Higuchi & Sakai, 2001; Lee, Kim et al., 2018; Marouf et al., 2012). The lack of consideration of additional logic is one of the major reasons why many previous studies cannot be applied to actual vehicles.

Finally, the system stabilization problem and the steering feel design problem are coupled to each other in the conventional EPS controller structure. Significant changes in the shape of the torque map can degrade the stability of the system because factors such as the slope of the torque map are considered in the design of the stabilizing logic. Therefore, each time the torque map is modified, it is necessary to re-analyze the system stability and modify the stabilization logic. Likewise, because stabilization logic generally consists of a combination of filters, it induces a phase shift, which distorts the steering feel determined by the torque map. Therefore, changes in stabilization logic parameters can have different effects on the steering feel each time. This coupling problem causes numerous trials and errors in the development process of the EPS controller, and when additional logic is considered, the process becomes more complex and time consuming.

To overcome the structural limitations of the conventional EPS controller mentioned above, the nonlinear torque map and additional logic that defines the steering feel and the stabilization logic that determines the performance of the controller must be separated from each other.

Therefore, in this study, the concept and design methodology of a controller that separates and simplifies the steering feel design problem and the system stability and control performance problem are proposed. The suggested controller, which is called the *target steering torque feedback controller*, consists of two separated modules: the *steering feel generation logic* module defines the target steering torque that the driver should feel under a given driving condition, and the *steering torque feedback controller* module controls the assist motor so that the driver actually feels the defined target torque, as shown in Fig. 3.

There have been a few studies that propose controllers that are conceptually similar to the target steering torque feedback controller.Kim and Song (2002) first proposed the concept of an EPS controller in which a target steering torque is determined and the driver's torque follows it, but a simple PI controller is utilized as a feedback controller. Williams (2009) proposed the concept of an add-on product for the hydraulic power steering system of the existing vehicle that modified the steering feel with additional assist torque from attached motor on steering column, but did not mention the design method of the controller. Lee, Yi, Chang, Lee and Jang (2018) further developed the study of Kim and Song (2002) and proposed a torque feedback controller based on an adaptive sliding mode controller, but the controller was designed based on a one degree of freedom EPS model with assumption that the EPS system is a single rigid body without compliance. Na, Li, Qiu, and Zhang (2020) conducted a study on the improvement of torque control performance degradation problem of motor driver due to the motor internal parameter change and back EMF in order to accurately control the assist motor of EPS system. The feedback controllers proposed in previous studies did not sufficiently reflect EPS system dynamics to be applied to a real vehicle, so there was a limit to ensuring stability and tracking performance. From the perspective of steering feel generation logic, previous studies commonly suggested a method of generating target steering torque based on the steering angle, steering angular velocity, but the steering feel generated in this way has limitation that it cannot reflect the information on the friction or roughness of the road surface. Lastly, controller performance was validated only in a simulation or hardware-in-the-loop simulation environment, not in actual vehicle experiments.

Section 2 explains how the proposed controller can overcome the limitations of the conventional EPS controller in detail. In Section 3, a design method for the steering torque feedback controller and the steering feel generation logic are proposed. A model-based linear torque feedback controller is designed in the frequency domain to stabilize the system with robustness against disturbances and system uncertainty and to follow the given target torque quickly and accurately. In addition, a design method of rack-force-based steering feel generation logic that can deliver a steering feel that is equivalent to that of conventional EPS controller and also can deliver road condition information to the driver is described. Finally, in Section 4, the proposed controller and design method are validated via simulation and actual vehicle experiments, respectively.

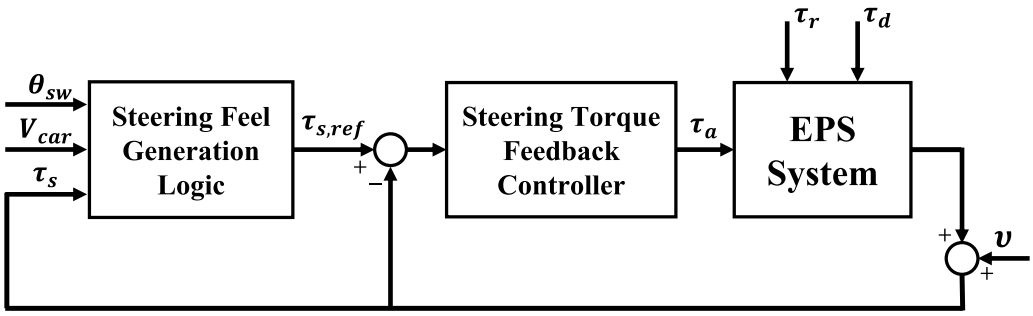


Fig. 3. Structure of a target steering torque feedback controller.

2. Target steering torque feedback controller

As shown in Fig. 3, the steering feel generation logic calculates the target steering torque ($\tau_{s,ref}$) that the driver should feel in a given situation by using vehicle and EPS state information such as vehicle speed, steering angle, and steering torque. The steering torque feedback controller controls the assist motor torque (τ_a) so that the sensed steering torque (τ_s) (or simply steering torque in this paper), can track the given target torque ($\tau_{s,ref}$) quickly and accurately.

Since the target steering torque feedback controller separates the driver's steering feel issue from the motor control and system stabilization problem, the controller can overcome the structural limitations of the conventional EPS controller mentioned in the introduction.

First, since the nonlinear torque map and additional logic for steering feel tuning are not included in the feedback controller, it is possible to analyze the system stability and to design a controller using linear control theory without linear approximation. Thus, the proposed controller is very practical compared to previous research with conventional EPS controller structures because previous controllers were based on a linearly approximated system and the effects of additional logic on system stability and performance were ignored. In addition, due to the characteristics of the EPS system, which requires high robustness to ensure safety and a high level of vibration suppression at the steering wheel, it is advantageous to apply linear control theory without approximation to ensure robust stability and tracking performance in the desired frequency range under system uncertainty.

Second, the steering feel tuning and a controller design ensuring the stability and performance of the system can be decoupled. For tuning the steering characteristics, the parameter or structure of the steering feel generation logic is changed, which results in changing only the target steering torque value but not the performance of the steering torque feedback controller. Conversely, any change in the steering torque feedback controller only affects to the performance of delivering the designed steering feel to the driver, not the designed steering feel itself. With the suggested structure of the EPS controller, once the feedback controller is designed to ensure the stability and performance of the system in advance, the steering feel can be freely designed based on the feedback controller environment.

Finally, the suggested controller allows a more intuitive steering feel design without limitations because the steering feel generation logic directly determines the target steering torque delivered to the driver. A conventional EPS controller has difficulty of knowing exactly how the torque felt by the driver would be affected when the shape of the torque map is changed or additional logic is added. However, the proposed controller always calculates the explicit value of the torque felt by the driver in a given situation. In addition, it is possible to define a new concept of steering feel that is completely free from the torque map + additional logic structure. For example, in a conventional EPS controller, it is impossible to design the steering feel so that the driver can feel zero torque from the steering wheel while driving because the torque map needs to have an infinite slope. However, in the proposed method, it is sufficient to simply set the target steering torque to zero.

Table 1
Model parameters and variables.

Notation	*Description*	*Notation*	*Description*
J_{sw}	Moment of inertia of steering wheel	J_{eq}	Equivalent moment of inertia of lumped mass
C_{sw}	Damping coefficient of steering wheel	C_{eq}	Equivalent damping coefficient of lumped mass
θ_{sw}	Steering wheel angle	θ_c	Column angle
K	Torsional spring constant	τ_d	Driver torque
τ_s	Sensed steering torque	F_r	Rack force
τ_m	Motor assist torque	τ_r	Equivalent rack force
τ_a	Equivalent motor assist torque	r_p	Pinion radius
N	Motor gear ratio	v	Sensor noise

3. Controller design

3.1. Steering torque feedback controller design

This chapter describes how to design a linear torque feedback controller by suggesting a loop-shaping criteria for the controller in the frequency domain.

The purpose of the controller is to ensure the stability of the system and allow the sensed steering torque (τ_s), which is an approximated value of driver torque (τ_d), to follow the target steering torque ($\tau_{s,ref}$) quickly and accurately while suppressing undesired vibrations of the system by controlling the motor assist torque (τ_a).

To design the steering torque feedback controller based on the model, the two degree of freedom EPS model is used, and the equation of motion is expressed as follows (Lee, Kim et al., 2018).

$$J_{sw}\ddot{\theta}_{sw} + C_{sw}\dot{\theta}_{sw} + K(\theta_{sw} - \theta_c) = \tau_d \tag{1}$$

$$J_{eq}\ddot{\theta}_c + C_{eq}\dot{\theta}_c + K(\theta_c - \theta_{sw}) = \tau_a - \tau_r \tag{2}$$

$$\tau_s = K(\theta_{sw} - \theta_c) \tag{3}$$

$$\tau_r = r_p F_r \tag{4}$$

$$\tau_a = N\tau_m \tag{5}$$

All variables and parameters used in EPS modeling are described in Table 1. The lumped mass includes all mechanical components of the EPS below the torque sensor such as the column, motor, and rack. According to Eq. (3), the sensed steering torque in the EPS system can be represented with the torsional spring constant of the torsion bar inside the torque sensor. The rack force(F_r) in Eq. (4) indicates the reaction force between the tire contact patch and the road surface that is transmitted through the rack.

To analyze the influence of the system inputs (i.e. driver torque, rack force, and assist torque) on the torque sensor value, Eqs. (1) and (2) undergo Laplace transformation and are merged to represent the transfer function as follows:

$$\tau_s = K(\theta_{sw} - \theta_c) = \frac{K(J_{eq}s+C_{eq})\tau_d - K(J_{sw}s+C_{sw})\tau_a + K(J_{sw}s+C_{sw})\tau_r}{J_{sw}J_{eq}s^3 + (J_{sw}C_{eq}+J_{eq}C_{sw})s^2 + (KJ_{sw}+KJ_{eq}+C_{sw}C_{eq})s + K(C_{sw}+C_{eq})} \tag{6}$$

Eq. (6) can be simplified as follows:

$$\tau_s = G_1(s)\tau_d - G_2(s)\tau_a + G_2(s)\tau_r \tag{7}$$

where $G_1(s) = \frac{K(J_{eq}s+C_{eq})}{\Delta}$, $G_2(s) = \frac{K(J_{sw}s+C_{sw})}{\Delta}$,

$\Delta = J_{sw}J_{eq}s^3 + (J_{sw}C_{eq} + J_{eq}C_{sw})s^2 + (KJ_{sw} + KJ_{eq} + C_{sw}C_{eq})s + K(C_{sw} + C_{eq})$

τ_a can be represented by the control error, sensor noise and $H(s)$, which is the Laplace transformation of the steering torque feedback controller, as shown in Eq. (8).

$$\tau_a = -H(s)(\tau_{s.ref} - \tau_s + \upsilon) \tag{8}$$

If the torque sensor's value is larger than the target steering torque value ($\tau_s > \tau_{s.ref}$), then the motor torque should be increased to reduce the torque that the driver feels. For this reason, Eq. (8) includes the negative feedback gain. By substituting Eq. (8) into Eq. (7), the transfer function from the driver torque, target steering torque, sensor noise and disturbance(rack force) to the torque sensor value can be described as follows.

$$\tau_s = \frac{G_1(s)}{1+G_2(s)H(s)}\tau_d + \frac{G_2(s)H(s)}{1+G_2(s)H(s)}(\tau_{s.ref} + \upsilon) + \frac{G_2(s)}{1+G_2(s)H(s)}\tau_r \tag{9}$$

To guarantee tracking performance, disturbance and noise attenuation and system stability, three loop-shaping criteria of the controller in the frequency domain are suggested as follows.

Criterion (1) $G_2(s)H(s)$ must have a large magnitude in the low-frequency range in which the target steering torque and disturbances are formed.

In the low-frequency range in which the target steering torque and disturbance are formed, the controller should be designed to satisfy Eq. (10) so that the torque sensor value is close to the target torque, and the influence of disturbance that includes the driver torque and rack force is minimized.

$$\frac{G_2(s)H(s)}{1+G_2(s)H(s)} \approx 1,\ \frac{G_1(s)}{1+G_2(s)H(s)} \ll 1,\ \frac{G_2(s)}{1+G_2(s)H(s)} \ll 1 \tag{10}$$

If the controller is designed to satisfy $|G_2(s)H(s)| \gg 1$, then the transfer functions can be approximated as follows:

$$\left|\frac{G_2(s)H(s)}{1+G_2(s)H(s)}\right| \approx 1,\ \left|\frac{G_1(s)}{1+G_2(s)H(s)}\right| \approx \left|\frac{G_1(s)}{G_2(s)H(s)}\right|,$$
$$\left|\frac{G_2(s)}{1+G_2(s)H(s)}\right| \approx \left|\frac{G_2(s)}{G_2(s)H(s)}\right| \tag{11}$$

As Eq. (11) shows, if the controller is loop-shaped to satisfy $|G_2(s)H(s)| \gg 1$ in the low-frequency range, then the target torque tracking and disturbance suppression performance can be obtained. Generally, if a tracking error less than ϵ is necessary, the controller should be designed to have a magnitude higher than the lower bound of $M = 1/\epsilon$ (for example, $M = 100$ when $\epsilon = 1\%$) in the frequency lower than maximum target steering torque frequency, as shown in Eq. (12).

$$|G_2(s)H(s)|_{s=j\omega} > M \gg 1 \ for\ \omega < \omega_{\tau_{s,ref},max} \tag{12}$$

Criterion (2) $G_2(s)H(s)$ should have a small magnitude in the high-frequency range in which the noise signals are formed.

If the designed controller satisfies $|G_2(s)H(s)| \ll 1$ in the high-frequency range, then the following is satisfied:

$$\left|\frac{G_2(s)H(s)}{1+G_2(s)H(s)}\right| \approx |G_2(s)H(s)| \ll 1 \tag{13}$$

Therefore, if Criterion 2 is satisfied with an upper bound of $m \ll 1$ higher than the minimum frequency of noise as shown in Eq. (14), noise suppression by the factor m can be achieved.

$$|G_2(s)H(s)|_{s=j\omega} < m \ll 1 \ for\ \omega > \omega_{v,min} \tag{14}$$

Criterion (3) The gain margin and phase margin of $G_2(s)H(s)$ must be positive and have sufficient value.

In addition to tracking and noise attenuation performance, it is also necessary to ensure the stability of the EPS system. To stabilize the system, a positive gain margin and a positive phase margin of the open loop transfer function of the feedback system($G_2(s)H(s)$) are necessary (Natarajan, 2005).

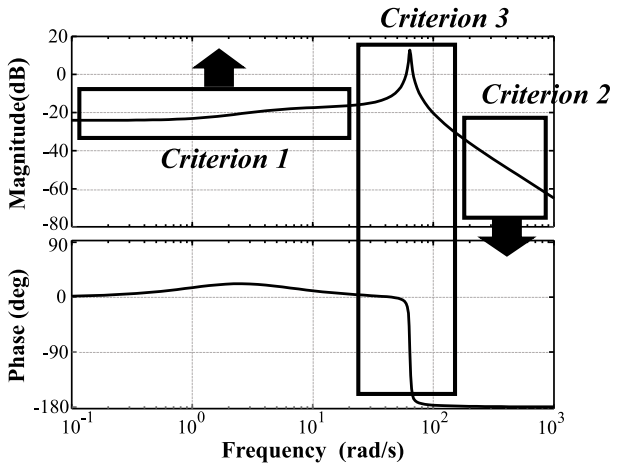


Fig. 4. Bode plot of $G_2(s)$ and loop-shaping guideline for a controller in the frequency domain.

In addition, sufficiently high gain margins and phase margins allow the system to be robust against system modeling uncertainty. In general, a phase margin of 45 degrees or more and a gain margin of 6 dB or more are enough to ensure that the stability of the system is robust (Willy, 2006).

In this study, the lead–lag compensator is used as a feedback controller to design a linear controller $H(s)$ that satisfies the suggested criteria. The Nth order lead–lag compensator is expressed as in Eq. (15).

$$H(s) = K_{LL}\frac{\frac{s}{z_1}+1}{\frac{s}{p_1}+1}\frac{\frac{s}{z_2}+1}{\frac{s}{p_2}+1}\cdots\frac{\frac{s}{z_n}+1}{\frac{s}{p_n}+1} \tag{15}$$

The lead–lag compensator can easily modify the gain and phase in the desired frequency range (Ogata & Yang, 2002; Wang, 2003). Because the lead–lag compensator has the same order of the numerator and denominator, the phase is only affected in the frequency band between the pole(p_i) and zero(z_i) of each lead–lag compensator. In addition, if the lead–lag compensator is defined as Eq. (15), the low frequency gain is affected only by K_{LL}, so the low frequency gain can be tuned without the influence of the phase adjustment parameters(p_i, z_i).

The magnitude of the lead–lag compensator is adjusted by the gain K_{LL}. The phase characteristic of the lead–lag compensator is determined by the values of the poles(p_i) and zeros(z_i). The case of $0<z_i<p_i$ is called the lead compensator, which raises the phase of the frequency band between the pole and zero but increases the high-frequency gain. In comparison, the case of $0<p_i<z_i$ is called a lag compensator, which reduces the high-frequency gain but loses the phase in the frequency band between zero and the pole.

In general, to satisfy the feedback controller design condition, the magnitude parameter K_{LL} is increased to increase the overall gain to satisfy Criterion 1. The magnitude in the high-frequency range is reduced by using the lag compensator to satisfy Criterion 2. To satisfy Criterion 3, loop-shaping with a lead compensator is applied to secure the phase margin and gain margin near the crossover frequency. To obtain a controller with the desired performance, it is necessary to perform fine tuning to satisfy the criteria while checking the Bode plot of $G_2(s)H(s)$ in the frequency domain. The total loop-shaping guideline in the frequency domain is represented in the Bode plot of $G_2(s)$ (when $H(s) = 1$) as shown in Fig. 4.

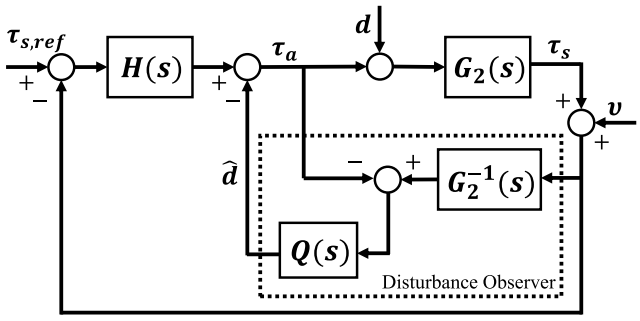


Fig. 5. Steering torque feedback controller with a disturbance observer.

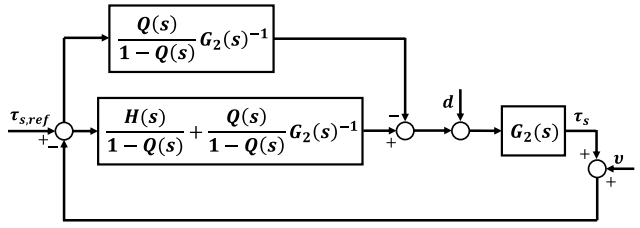


Fig. 6. Rearranged block diagram of feedback controller with the Q filter.

3.1.1. Disturbance observer

One of the important factors that determines the performance of the feedback controller is how well it suppresses the effect of disturbance. If the input and output of the system plant are set to motor assist torque (τ_a) and sensed steering torque (τ_s), and the effects of the rack force(τ_r) and driver torque (τ_d) on the system are lumped as one disturbance term, the lumped disturbance d is defined as Eq. (16) and system block diagram is rearranged as Fig. 5 based on Eq. (7).

$$d = \tau_r + G_2^{-1} G_1 \tau_d \tag{16}$$

Since the rack force generally has a very large value relative to the target steering torque, especially in the parking state, it is difficult to obtain the desired tracking performance via disturbance suppression of the feedback controller alone. Therefore, the disturbance observer(DOB) is utilized to prevent the performance degradation of the feedback controller due to large disturbances.

The block diagram in Fig. 5 shows how the feedback system with the DOB works. The estimated disturbance($\hat{d}$) can be calculated by subtracting the control input (τ_a) from the product of the measurement ($\tau_s + v$) and the inverse system(G_2^{-1}). However, the inverse system(G_2^{-1}) generally has a causality problem because the degree of the numerator is larger than that of the denominator. To solve this problem, a low-pass filter $Q(s)$ called the *Q-filter* is applied as given by Eq. (17) (Chen, Yang, et al., 2016; Choi, Yang, Chung, Kim, & Suh, 2003).

$$\hat{d} = Q(s)(G_2^{-1}(\tau_s + v) - \tau_a) \tag{17}$$

By subtracting the estimated disturbance($\hat{d}$) in the control input of the feedback controller, the effect of actual disturbance to the feedback system can be compensated.

The second-order low-pass Q-filter with cutoff frequency (ω_{DOB}) is used in this study as shown in Eq. (18) because the difference between the denominator and the numerator of $G_2(s)$ is 2.

$$Q(s) = \frac{\omega_{DOB}^2}{(s + \omega_{DOB})^2} \tag{18}$$

If the block diagram in Fig. 5 is rearranged as shown in Fig. 6, then the effect of the DOB on the feedback system can be divided into feedforward and feedback components. As shown in Fig. 6, adding a DOB has the effect of adding a feedforward controller that includes

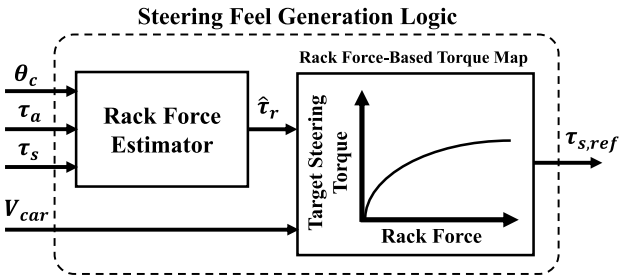


Fig. 7. Configuration of steering feel generation logic.

system model information and improves the reference tracking performance. In addition, the feedback system with the DOB has a superseded controller $H(s)^* = \frac{H(s)+Q(s)G(s)_2^{-1}}{1-Q(s)}$ instead of $H(s)$. Therefore, a new transfer function $G_2(s)H(s)^*$ can be used to design a feedback controller that satisfies the proposed criteria from the previous chapter.

$$G_2(s)H(s)^* = \frac{G_2(s)H(s) + Q(s)}{1 - Q(s)} \tag{19}$$

The complexity of the $G_2(s)H(s)^*$ makes it difficult to intuitively perform loop-shaping to achieve the desired magnitude and phase. However, when analyzing the effect of applying the DOB on the Bode plot of $G_2(s)H(s)^*$, the influence of $Q(s)$ can be neglected and $G_2(s)H(s)^*$ can be approximated as $G_2(s)H(s)$ at a frequency above ω_{DOB} because of the low-pass-filter characteristics of the $Q(s)$. In addition, at a frequency below ω_{DOB}, the DOB raises the low-frequency gain of $G_2(s)H(s)^*$ to infinity and this increased gain improves the tracking and disturbance suppression performance of the controller; however, it can cause system instability due to the phase delay. Therefore, applying a Q-filter with an appropriate cutoff frequency to a predesigned $G_2(s)H(s)$ satisfying the controller design criteria can improve the tracking performance without affecting the stability of the system. This appropriate cutoff frequency should be lower than the frequency at which the phase margin and gain margin of $G_2(s)H(s)$ are formed and higher than the frequency that contains the disturbance to be attenuated.

3.2. Steering feel generation logic

Although various implementation methods can be used to create the steering feel generation logic, A method using an external rack-force-based torque map is proposed in this paper. Fig. 7 shows the structure of the rack-force-based steering feel generation logic.

The rack force estimator calculates the rack force based on EPS system states, and the torque map determines the target steering torque based on the estimated rack force. There are advantages to designing the steering feel using a rack-force-based torque map.

First, the rack-force-based steering feel generation logic can deliver road condition information because the rack force is a raw signal that contains all information transmitted from the road surface to the tire. The rack force includes the self-alignment torque and road friction, which reflect whether the road is dry, wet or frozen and vibration characteristics that reflect the roughness of the road. The role of the torque map in Fig. 7 determines the ratio between the driver's steering torque and rack force. The driver will always feel a certain portion of the rack force, allowing the driver to detect changes in road conditions. If the proper filter is added before or after the torque map, then the roughness of the road surface included in the rack force can be properly processed and transmitted to the driver.

Furthermore, rack-force-based torque map can provide a steering feel similar to the steering feel of a conventional EPS controller that is familiar to drivers. Eq. (20) represents the equation of motion of the EPS system when the system is assumed to be one rigid body.

$$\tau_d + \tau_a = \tau_r + J\ddot{\theta} + C\dot{\theta} \tag{20}$$

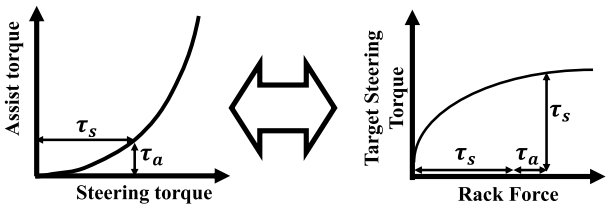


Fig. 8. Relation between the torque map of a conventional EPS controller and the rack force-based torque map.

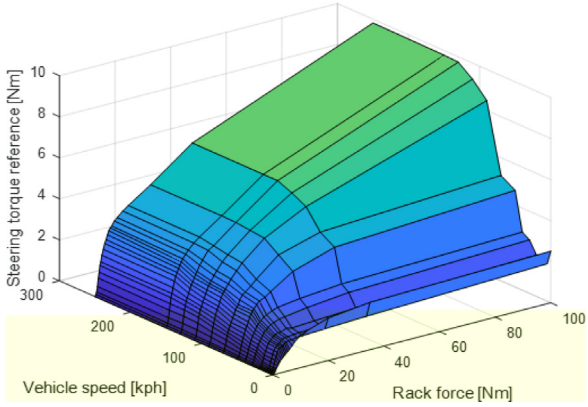


Fig. 9. Steering torque map using the rack force.

If the system is in a steady state ($\ddot{\theta} = \dot{\theta} = 0$), then the driver torque, motor torque and rack force are balanced, as shown in Eq. (21).

$$\tau_d + \tau_a = \tau_r \tag{21}$$

After the rack force is determined by the road condition, vehicle speed and steering angle, the controller determines only the ratio between the assist torque of the motor and the torque felt by the driver according to Eq. (21). In other words, the proposed torque map between the rack force and target steering torque and the torque map of the conventional EPS controller between the driver torque and assist torque can be transformed. The transformation from the torque map of the commercialized passenger car with the conventional EPS controller to a rack-force-based torque map is described in Fig. 8.

The typical shapes of the rack-force-based target steering torque map and the torque map of conventional EPS controller are shown in Fig. 8. The rack-force-based torque map has a steep slope near the origin and becomes gradual as the rack force increases due to the typical bell-shaped torque map of conventional EPS controller. Large gradient of the rack-force-based torque map near the origin makes it easy for the driver to detect the neutral position of steering wheel.

Fig. 9 shows the 3-dimensional torque map varying with the vehicle speed. At low vehicle speed condition including frequent cornering and large steering angles, comfortable steering is possible with less target steering torque. When the vehicle speed is high, the steering feel needs to become more firm by applying the proper slope of the torque map to improve driving stability.

The rack force can be estimated based on the EPS model as given by Eq. (22).

$$\hat{\tau}_r = \tau_a + \tau_s - J_{eq}\ddot{\theta}_c - C_{eq}\dot{\theta}_c \tag{22}$$

The $\dot{\theta}_c$, $\ddot{\theta}_c$, τ_s, and τ_a are signals that can be obtained from the EPS ECU(Electric Control Unit); thus, the rack force can be calculated directly using Eq. (22).

The steering feel generation logic that realizes a steering feel similar to the conventional EPS controller proposed in this paper is an example or starting point of designing the steering feel. It is expected that future studies on steering feel design and evaluation can be carried out based on the proposed controller structure.

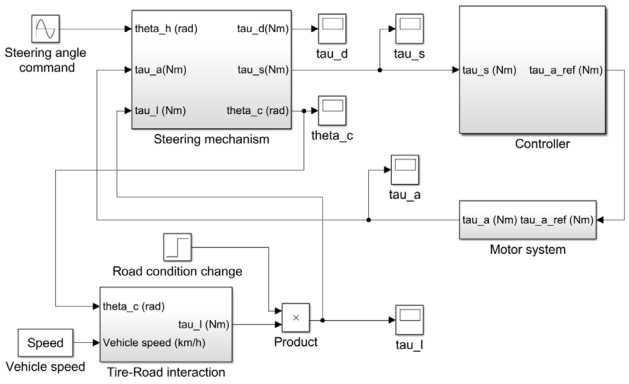


Fig. 10. Simulink simulation of the EPS system.

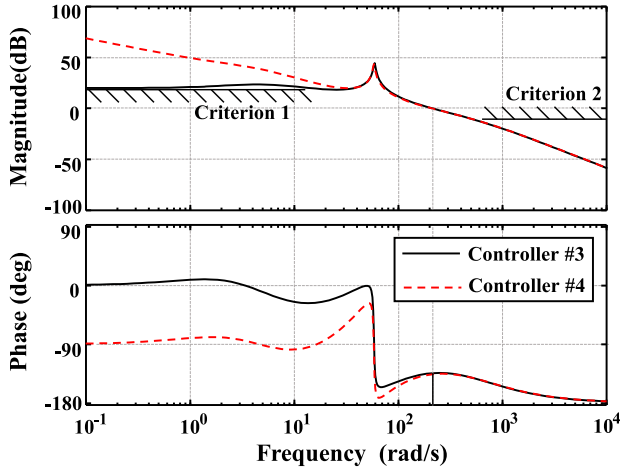


Fig. 11. Bode plot of the open loop transfer function of Controller #3 and #4.

Table 2
Simulation parameter.

Parameter	Value	
J_{sw}	0.0412	[N m^2]
C_{sw}	0.0691	[N m s/rad]
J_{eq}	0.253	[N m^2]
C_{eq}	1.03	[N m s/rad]
K	144	[N m/rad]
$Friction$	2	[N m]
ω_m	50	[Hz]
$K_{tire,park}$	30	[N m/rad]
M	20	[dB]
m	−12	[dB]
$\omega_{\tau_{s,ref},max}$	2	[Hz]
$\omega_{v,min}$	100	[Hz]

4. Simulation and vehicle test result

4.1. Simulation environment

To verify the performance of the proposed target steering torque feedback controller, a simulation environment was established using MATLAB/Simulink as shown in Fig. 10.

The simulation consists of the EPS mechanism, controller, motor dynamics, and road disturbance block. The EPS mechanism block is based on Eqs. (1)–(4), and coulomb friction has been added to the lumped mass to achieve behavior similar to a real system. The steering torque

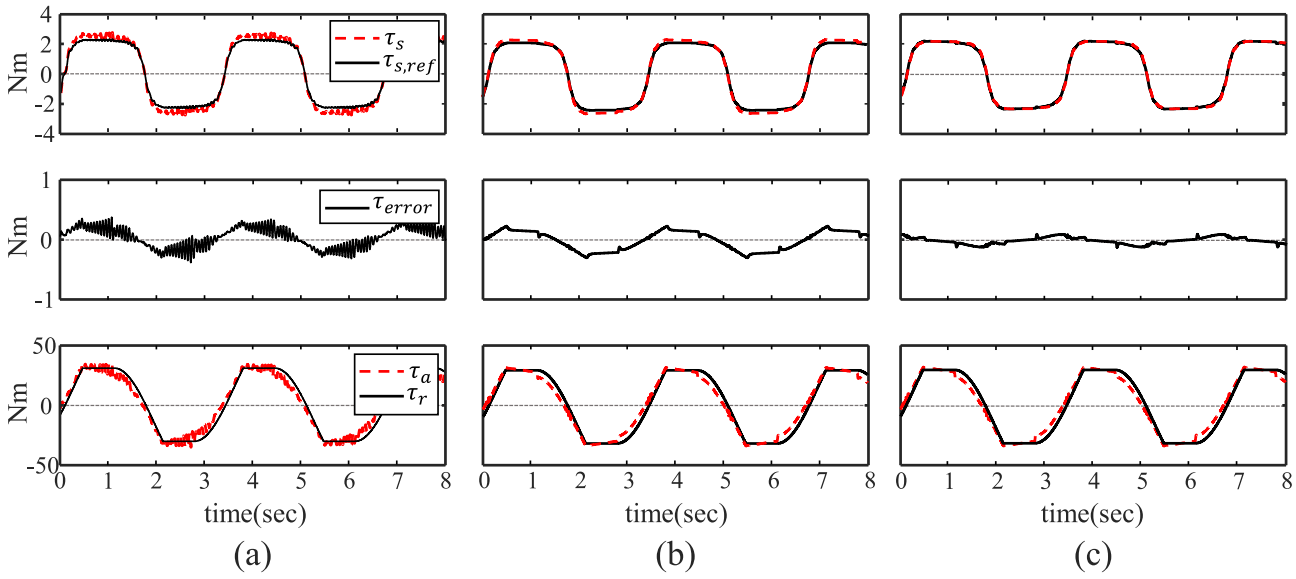


Fig. 12. Simulation results with Controller # (a) 2, (b) 3, and (c) 4.

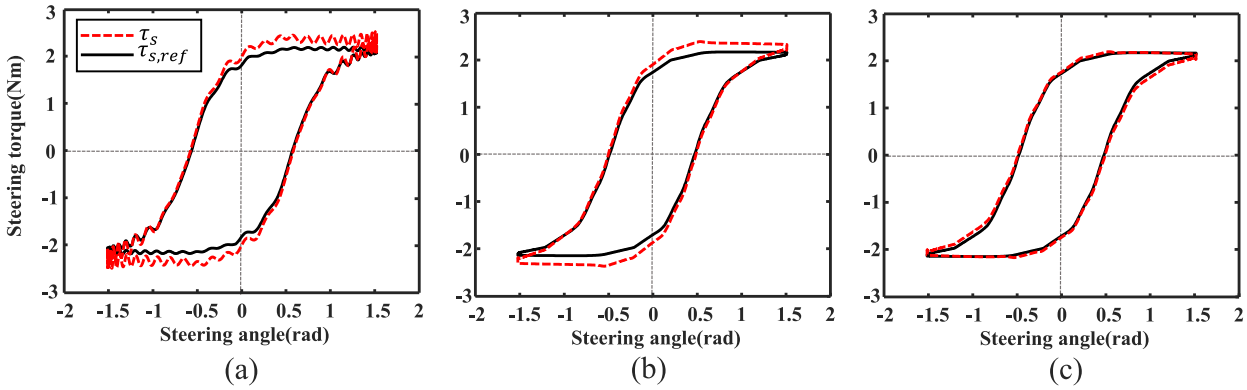


Fig. 13. Hysteresis curve comparison with the reference torque and measured torque for Controller # (a) 2, (b) 3, and (c) 4.

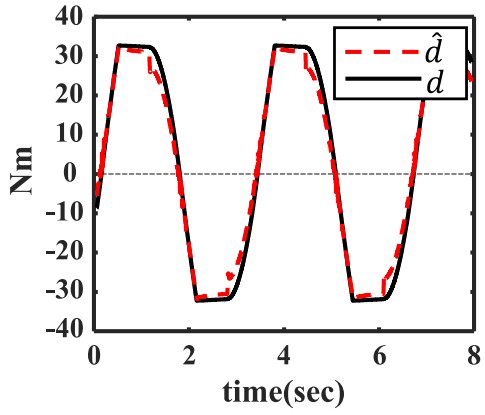


Fig. 14. Disturbance and estimation value.

feedback controller with the DOB and a steering feel generation logic block are based on Figs. 5, 7 and 9. A first-order low-pass filter with cutoff frequency (ω_m) is used to express the response characteristics of the motor torque control system. The road disturbance, which is the interactive moment between the tire and the road surface, is divided into two modes: the parking state and the driving state. In the driving state, the road disturbance is proportional to the column angle ($\theta_{c,k}$), as shown in Eq. (23). In the parking state, the road disturbance is represented as the saturation function with hysteresis given by Eq. (24).

$$\tau_{r,k} = K_{tire,drive}\theta_{c,k} \tag{23}$$

$$\tau_{r,k} = K_{tire,park}(\theta_{c,k} - \theta_{t,k}) \tag{24}$$

$$\theta_{t,k} = \begin{cases} \theta_{c,k} - \theta_{max}, if\, \theta_{c,k} - \theta_{t,k-1} > \theta_{max} \\ \theta_{c,k} + \theta_{max}, if\, \theta_{c,k} - \theta_{t,k-1} < -\theta_{max} \\ \theta_{t,k-1}, otherwise \end{cases} \tag{25}$$

$\theta_{t,k}$ indicates the equivalent angular position of the tire contact patch at the discrete time step k and represents the actual slip of the tire contact patch excluding the tire deformation from the steering angle.

The values of system parameters used in the simulation and controller design parameters are listed in Table 2. The system parameters in Table 2 are obtained through the identification process with the actual EPS system of a vehicle for experiment.

4.2. Simulation result

Four different controller parameter sets presented in Table 3 are selected to show the performance difference of the system according to the feedback controller parameter design.

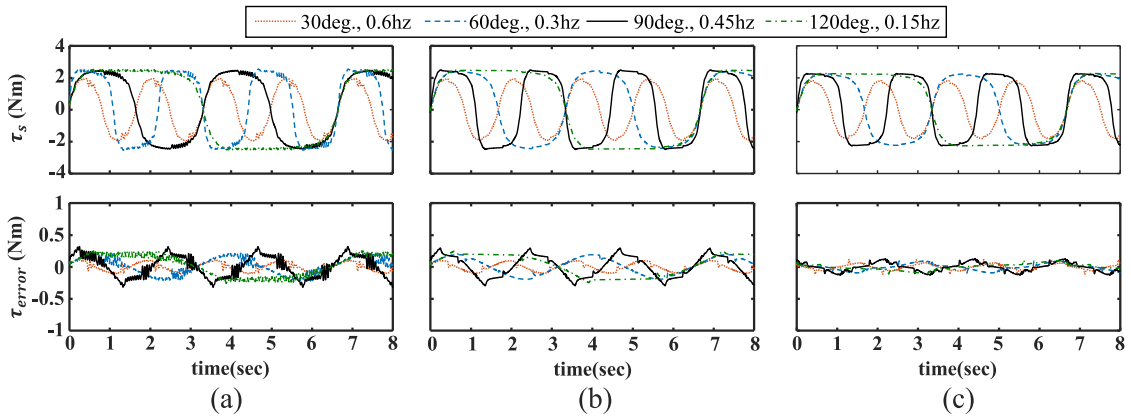


Fig. 15. Simulation results with various steering angle input with Controller # (a) 2, (b) 3, and (c) 4.

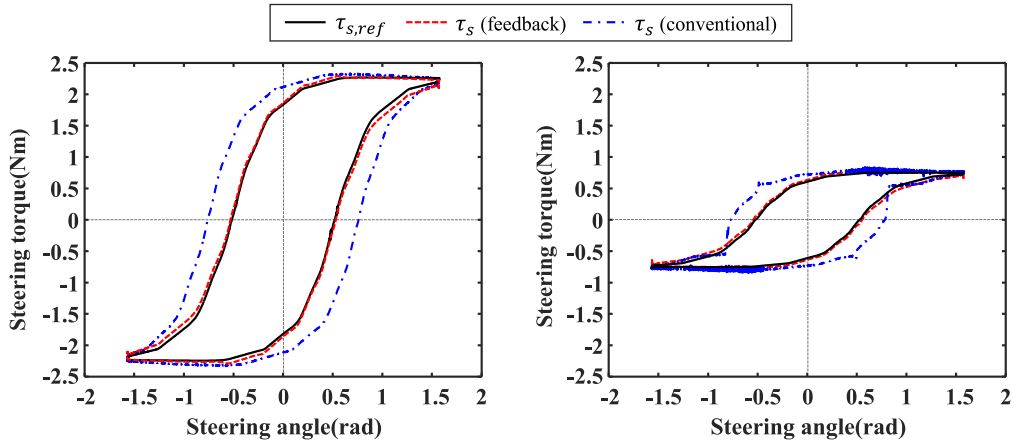


Fig. 16. Comparison between the reference torque and measured steering torque from system with conventional EPS controller and target steering torque feedback controller with (a) original torque map and (b) the modified torque map (with 1/3 times decreased steering torque for driver).

Table 3
Pole, Zero, and DOB parameters and controller performances of various controller cases.

Controller	1	2	3	4
K_{LL}	160	160	160	160
p_1	6	6	6	6
z_1	35	35	35	35
p_2	1	120	110	110
z_2	1	65	65	65
p_3	1	1	600	620
z_3	1	1	115	115
ω_{DOB}	N/A	N/A	N/A	50 rad/s
GM	−33.4 dB	Inf	Inf	Inf
PM	−9.59 deg.	3.84 deg.	46.2 deg.	45.9 deg.

The lead–lag compensator gain (K_{LL}) is large enough to satisfy Criterion 1 for tracking performance. The parameters of controllers 1 and 2 were selected to show the performance degradation as a result of poorly selected controller parameters. Controller 1 shows the case with a negative phase margin and a negative gain margin and Controller 2 shows the case with a very low positive phase margin. Controller 3 was designed to meet all suggested criteria for the performance and stability of the feedback controller using a third-order lead–lag compensator based on the loop-shaping guideline of Fig. 4. Controller 4 adds a DOB to controller 3, and the DOB cutoff frequency is selected high enough to estimate the driver torque and rack force but low enough to avoid the influence on the gain and phase margin of the system. Table 3 and Fig. 11 show that Controllers 3 and 4 satisfy all the design criteria.

The simulation was conducted in the parking state, which is the worst condition in terms of system stability because of the large and nonlinear road disturbance, and the input command of the simulation is the steering wheel angle with a sinusoidal waveform that has a frequency of 0.3 Hz and an amplitude of 90 degree.

The simulation result applying Controller #1 shows that when the controller does not satisfy the stability condition (Criterion 3), the system is unstable and divergent. The simulation result with system divergence is not graphed. The system with Controller 2 does not diverge, as shown in Fig. 12(a), because Controller #2 satisfies the positive phase margin and positive gain margin criterion. However, Controller #2 causes a critical level of vibration in the system. This result shows that a controller without robust stability can cause undesired vibrations when there is parameter uncertainty or unmodeled dynamics. In the simulation, friction force between gears is applied as an unmodeled nonlinear dynamics. Controller #3 has sufficient gain and phase margin(Criterion 3) and satisfies the criteria for tracking performance(Criterion 1) and noise attenuation(Criterion 2); thus, the system can track the target steering torque without undesirable vibration, as shown in Fig. 12(b). With Controller #3, the torque sensor value follows the target steering torque value with an error of 0.4 N m or less.

Controller #4 with an additional DOB improves the performance of the feedback controller to reduce the tracking error by half (<0.2 N m) by directly estimating and canceling the disturbance from the road surface, as shown in Fig. 12(c). Since the cutoff frequency of the DOB (ω_{DOB}) is selected for the DOB to follow the disturbance without affecting the stability of the feedback system, it is possible to improve the tracking performance as well as not to degrade the phase margin

and gain margin by applying the DOB to Controller #3, as shown in Fig. 11.

Fig. 13 shows the hysteresis curve between the steering angle and the steering torque to compare the steering feel produced by the steering feel generation logic with the actual steering feel. It can be seen that the result of the well-designed feedback controller with the DOB follows the target hysteresis curve accurately with a small error bound.

Fig. 14 shows that the designed DOB estimates the actual disturbance well.

Fig. 15 shows that the performance of the proposed controllers show the same tendency for various steering angle inputs with different frequencies and amplitudes. For Controller #2 with insufficient robustness, the vibrations were too severe for drivers to operate and for Controllers #3 and #4, the system stability is guaranteed with the various steering input. In addition, it can be seen that regardless of the steering input, the controllers guarantee a similar level of tracking error.

Fig. 16 compares the performance of the proposed feedback controller and conventional EPS controller. A controller structure of using a lead–lag compensator as a stabilizing filter (Lee, Kim et al., 2018) was utilized for the conventional EPS controller. A torque map of the conventional EPS controller is obtained by transforming the rack force-based torque map of Fig. 9 according to the Fig. 8. With the obtained torque map, the conventional EPS controller parameters are designed using the controller design method of Lee, Kim et al. (2018), but in order to have same noise reduction performance of suggested feedback controller, the high frequency attenuation conditions were added. The hysteresis curve of the measured steering torque and reference torque from the simulation with the conventional EPS controller and target steering torque feedback controller (with parameters of Controller #4 in Table 3) is compared, as shown in Fig. 16(a). Conventional EPS controller has relatively low tracking performance of the reference torque because stabilizing logic and system damping and inertial characteristics cause distortion to steering feel determined in the torque map.

In addition, the torque map is modified for the driver to feel only 1/3 of the steering torque without the changing controller parameters, and the simulation result is shown in Fig. 16(b). In the case of the target steering torque feedback controller, the feedback controller tracks the target steering feel well even after modifying the steering feel generation logic. However, in the case of the conventional EPS controller, a large change in the torque map causes the deterioration of tracking performance and undesirable vibration because the shape of the torque map itself affects the design of the stabilizing filter of controller. When the torque map is tuned to reduce the steering torque felt by the driver to less than 1/3, the simulation result of the conventional EPS controller is diverged, while the target steering torque feedback controller performs well.

In the process of modifying the torque map for reducing the steering torque, the characteristics of the two controllers can be compared. To make the driver feel 1/3 of the original steering torque, for the target steering torque feedback controller, the rack-force-based torque map was simply reduced by 1/3 on the y-axis. However, in the case of conventional EPS controllers, there is no explicit method to modify the torque map. Therefore, either the method of converting from a modified rack force-based torque map or that of obtaining a new torque map through a trial-and-error process can be considered.

4.3. Vehicle test result

The performance verification of the controller with a real vehicle (Hyundai motors, Genesis) with the identified EPS system parameters in Table 2 was carried out with various controllers, including a PID(proportional–integral–derivative) controller with a DOB, a lead–lag compensator with a DOB and an LQG(linear quadratic Gaussian) controller. The PID controller with a DOB substitutes $H(s)$ in Fig. 5 with the PID controller and the LQG controller is composed of a Kalman observer with unknown inputs that acts as a state observer and a DOB and a linear quadratic tracking controller (Catlin, 1989; Mohamed, Zasadzinski, Onana, & Nowakowski, 1995). The lead–lag compensator with a DOB has the same parameters as Controller 4 in Table 3, and the PID with a DOB and LQG are tuned to maximize tracking performance without producing high frequency vibration that exceeds a certain level (<0.2 N m). The repeated steering input and constant steering velocity input, as shown in Fig. 17, were applied in the parking state.

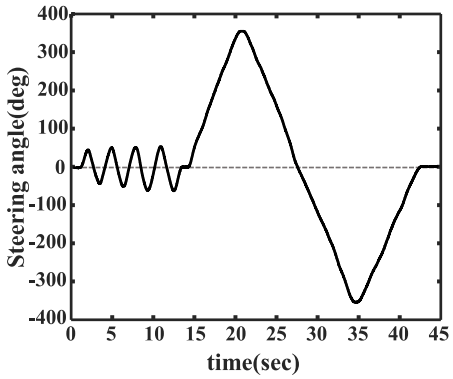


Fig. 17. Steering angle input for the vehicle test.

Table 4
Steering torque feedback controller tracking error from vehicle test with different type of controller.

	PID + DOB	LeadLag + DOB	LQG
Repeated steering angle input (peak value)	± 0.3 N m (0.5 N m)	± 0.1 N m (0.25 N m)	± 0.9 N m
Constant steering angular velocity input	>0.2 N m	<0.05 N m	>0.6 N m

The tracking error performance of different controllers is shown in Fig. 18 and Table 4. The experimental results show that the performance of the lead–lag compensator + DOB is the best and that the error is controlled within 0.1 N m. It can be seen that the magnitude and waveform of the error from the vehicle experiment are similar to the simulation result. The performance of the controller using the lead–lag compensator is better than that of the others because the lead–lag compensator has more degrees of freedom for loop-shaping in the frequency domain than the PID controller, and it is difficult to design the LQG in the frequency domain.

5. Conclusion

This study explains advantages of the target steering torque feedback controller compared to conventional EPS controllers in detail to justify the necessity of applying the proposed controller to real vehicles, and suggested a overall controller development process including a steering torque feedback controller design that allows the driver to experience the designed steering feel accurately and stably, and a steering feel generation logic design that enables an intuitive steering feel design. For the steering torque feedback controller, based on the EPS model, how to design the controller to make the system stable while achieving tracking performance and noise suppression through loop-shaping in the frequency domain is shown. In addition, adding a disturbance observer to the feedback controller minimizes the effects of disturbances and improves the performance of the feedback controller. For steering feel generation logic, a steering feel design method using a rack-force-based torque map that can create a steering feel similar to that of the conventional EPS controller is proposed. The simulations show that a controller that meets the proposed criteria ensures that the

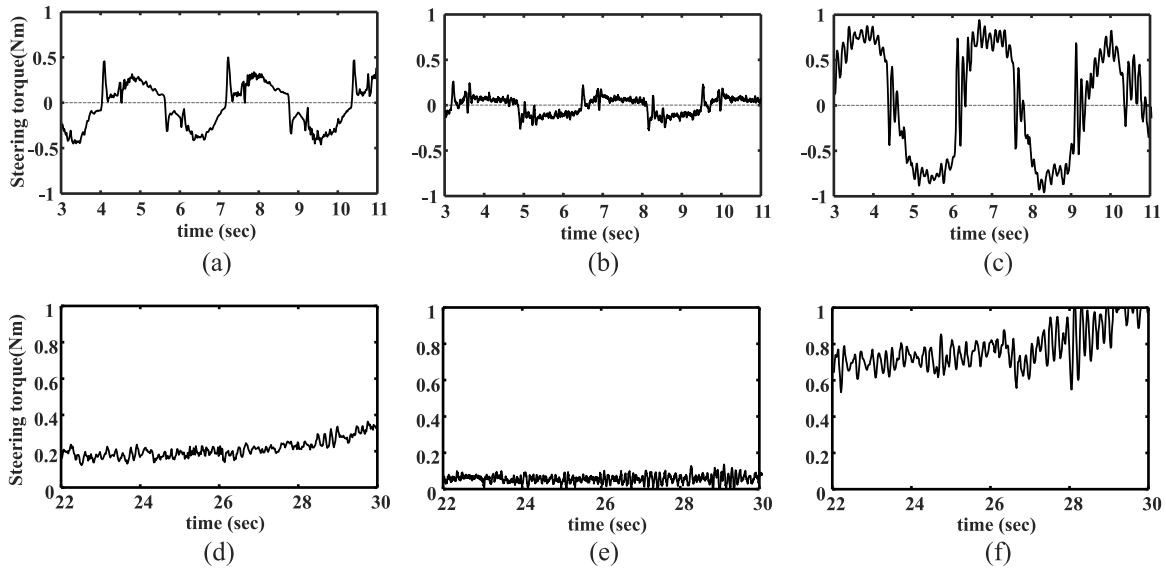


Fig. 18. Vehicle test results with (a,d) a PID with a DOB, (b,e) a Lead–lag compensator with a DOB and (c,f) an LQG controller, (Steering torque feedback controller tracking error from (a,b,c) repeated steering input, (d,e,f) constant steering velocity input).

system is stable and has sufficient target torque tracking performance. Finally, vehicle tests show that the controller verified in the simulation works well in a real system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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